The Statistical Grimoire: Modern Statistical Approaches for the Natural Sciences Using R

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About

The Statistical Grimoire is a beginner-friendly open-access textbook actively being updated with statistical and data analysis techniques using the R programming language and other open-source tools.

This book was created to give students a solid, practical, and free introduction to both statistics and R programming, especially if they have not done any coding before.

> **Caution**
>
> You're reading a what amounts to a friendly beta version (it literally only has one complete chapter so far). Check back for additions and improvements.

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https://github.com/statistical-grimoire/book

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- **Body Typeface:** Atkinson Hyperlegible Next
- **Code Typeface:** `Atkinson Hyperlegible Mono Font`

Why This Book Exists

Many students arrive at graduate level study with a strong dislike of statistics and, importantly, without the preparation modern research demands. The gap here isn't only a statistical one. It's also about the tools and workflows that make statistical work reliable, transparent, and reproducible.

Open-source tools such as R, Git, Markdown, and LaTeX, and many others are now standard in many research settings. Students paying for a rigorous education should be introduced to these tools early. Yet many never even learn that they exist, and those that do typically discover them by accident since most programs barely mention them. Those who do encounter them often learn under pressure, quickly and inconsistently, because a deadline leaves them with no alternative. That path is stressful and prone to mistakes. Its the route I and many others took and it left me frustrated and resentful. How could an overpriced education fail so badly at preparing students for the practical realities of contemporary research.

These tools are not distractions from learning statistics; they support the process. They help students compute, document, verify, and communicate their work with far less friction than would otherwise be experianced. In an era of unprecedented access to computing power, it makes little sense to train students as if that power, and the practices that harness it, barely matter. When we cling to older methods by default, we risk mistaking tradition for rigor and blaming students for problems our training choices create.

This book is an attempt to close that gap: to teach statistics alongside the open-source tools and workflows that students will actually use, so they can spend less time fighting the process and more time doing careful, credible research.

The primary tool used throughout this book is the R programming language, though we will introduce other tools (markdown, git, LaTeX) when appropriate. Over the long term, my goal is for this text to function as a practical, open-access guide that leads complete beginners step by step through both statistical thinking and statistical programming in a clear, thorough, and usable way. Please treat everything here as a work in progress: the book will evolve.

Foolish Assumptions Made by Your Author

- You will actually read this (yes, I'm an optimist)

- You probably don't like math. You might even fear it, but deep down, you know it's good for you. Math, like confession, is rarely pleasant but often necessary.
- You'll start at the beginning and move forward.
- You won't take the sillier stuff too seriously—especially the parts that sound true. They probably aren't. Probably.

Errata

No manuscript is free from error. Should you unearth any mistakes, please file an issue on GitHub:

https://github.com/statistical-grimoire/book/issues

1 Introduction to R

1.1 What is R?

R is a programming language. A progamming language is simply a language humans use to communicate instructions to a digital computer.

1.2 The Genesis of R

The name "R" was derived from the first initials of its original two programmers, "R"obert Gentleman and "R"oss Ihaka. The decision to name the language using a single English letter is what might, charitably, be called a joke on the part of these two programmers, who saw themselves poking fun at R's parent language, which was given the unimaginative name of "S". In the 1970s the S language had undergone its initial development at the famous Bell Laboratories with the primary aim of enabling and encouraging "GOOD DATA ANALYSIS"- a goal so fundamental to the ethos of S that the authors, Becker and Chamber (1984), felt they had to emphasize it using uppercase lettering inside the preface to the language's inaugural instruction manual (the uppercase lettering has been reproduced here for the reader's benefit). The familial correspondence R has with S is present even to this day, to such an extent that Becker and Chamber's original manual could probably function decently well as an introductory manual to R itself.

1.3 Why a Progamming Language?

At this point readers might be wondering why it should ever be necessary to learn a programming language to conduct statistics and data analysis more generally. These topics are usually considered difficult enough by many students and educators, what need is there to compound this with a programming language? Why not, for instance, make use of any one of the many pieces of statistical software that already exist and (as the advertising would want you to beleive) require no requisite knowledge of programming? In other words, why not use software

such as SPSS[1] or one of its many malformed, and equally expensive, doppelgängers, Minitab, SAS, and Stata.

The primary answer to this question lies in flexibility. There is rarely a single correct way to analyze data, as different datasets come with their own unique challenges and intricacies. These complexities often resist the rigid, prescriptive approaches employed by many proprietary software programs. This is not to say software like SPSS cannot adapt to such scenarios—it often can. However, this adaptation typically comes at a cost: users may need to pay for additional features not included in the original purchase, or they may face an even steeper learning curve. One that forces mastery of a obscure and enigmatic language. A language that is so specific to the software, that only a select few (if any) even seem to understand it.

In direct contrast to this, R offers an intuitive and empowering experience for users. While it may seem daunting at first, R operates in a straightforward and logical manner, much like a calculator. Many users discover that using R is far easier than they initially expected. This is largely due to the vibrant and dedicated R community that exists online, which has cultivated an extensive network of resources over the years. Acolytes of R see it as something worthwhile to preserve and develop (often at their own personal time and expense). Proprietary statistical software has no equivalent to this, nor will it ever. Users are often snared within its ecosystem not out of preference or love for the program, but because it is all they have ever known.

Although programs like SPSS may initially appear familiar to new users due to their resemblance to popular spreadsheet software like Microsoft Excel, this familiarity is purely superficial. To illustrate, SPSS and its equivalents will, in a manner similar to most spreadsheet software, present users with an array of buttons and menus above a friendly spreadsheet-style grid. However, the similarities end there. New users will quickly find themselves overwhelmed by the abundance of strange options needed to perform even simple tasks, such as loading and viewing a dataset. Contrary to what their marketing might lead you to believe, the learning curve for these programs is dangerously steep, and users are unknowingly at risk of being lead off a cliff.

Owing to its nature as a programming language crafted for statistics, R is a language grounded by the logic of mathematics. This foundation often makes it easier for new users to understand and build upon, even for those who claim to dislike math. Moreover, proficiency with R grants users the ability to work with other statistical software if needed. The reverse, however, is rarely true: mastering SPSS or similar programs does not provide the same level of flexibility or transferable skills.

An altogether different answer to the question that opened this section, and one that will appeal to the University students reading this, is simply cost. R is free for the user, with no need to put up with annoying advertising or pay for additional features. The same can not be said of the other aforementioned software which are almost always subscription based,

[1] SPSS is popular software for conducting statistics that was originally released in the late 1960s and is an acronym for Statistical Package for the Social Sciences. At some point it was purchased by IBM and re-branded to mean *Steeply Priced Shitty Software.*

requiring the user to consistently renew a very expensive license to use the software. But R is not just free in monetary terms, it is also free in philosophical terms. R adopts the Free Software Foundation's GNU General Public License and thus adheres to the philosophy of *free software* (what some might term *open-source*).

! The four essential freedoms

From the GNU project website (Free Software Foundation 2026):
A program is free software if the program's users have the four essential freedoms:

- The freedom to run the program as you wish, for any purpose (freedom 0).
- The freedom to study how the program works, and change it so it does your computing as you wish (freedom 1). Access to the source code is a precondition for this.
- The freedom to redistribute copies so you can help others (freedom 2).
- The freedom to distribute copies of your modified versions to others (freedom 3). By doing this you can give the whole community a chance to benefit from your changes. Access to the source code is a precondition for this.

This philosophy extends beyond the software itself to include both its file formats and help documentation. For years, the dissemination of scientific findings has been (and still is) hindered by the reliance on proprietary file formats imposed by commercial research tools. Locking information within these exclusive systems is clearly counterproductive to scientific progress, as it binds researchers to overpriced, branded ecosystems. Such practices prioritize profit over the broader goals of accessibility and collaboration, making their continued adoption ethically questionable. In practical terms, this means that choosing R is not just about its functionality, it is also a statement against the restrictive and exploitative behaviours proprietary software providers perpetuate. More plainly, and if for no other reason, we should use R just to give the middle finger to these companies.

As if you didn't need any other reasons to start using R immediately, here are some more:

- **R Is Not A Gooey Mess:** Unlike programs tied to a graphical user interface (GUI, often called "gooey"), R is not limited by point-and-click constraints. Its capabilities are as vast as what you and others can program (and your computer can handle).
- **Advanced Statistical Capabilities:** R's packages make it easy to apply best practices in statistics.
- **Enhanced Data Visualization:** With intuitive tools like *ggplot2*, *Shiny*, *Plotly*, and many others, R easily permits sophisticated and customized visualizations.
- **Reproducible Research:** R is built for reproducible research, aligning with open-science principles. It allows you to create scripts that are easy to share, review, and rerun, by anyone. This helps ensure transparency, accuracy, and reliability.
- **Integration with Other Tools:** R can easily integrate with other software and programming languages, such as Python, SQL, HTML, LaTeX, and even Excel. This

makes it a valuable tool for working in diverse computational environments. Moreover, because of it's nature as programing language, version control is seemlesly applied with systems like Git.

While the prospect of learning a programming language like R may seem daunting at first, it ultimately provides a more adaptable, intuitive, ethical, affordable, and rewarding tool for statistical analysis than many of its proprietary counterparts.

1.4 Why R?

At this point, it is worth addressing a question that comes up often: "Why use R instead of something like Python or Julia?" It is a fair question. After all, Python and Julia—like R—are full-fledged programming languages that are powerful and capable, but the difference comes down to origin story.

R is not a general-purpose language. It is the progeny of S, a language birthed in the depths of statistical practice for one primary goal: GOOD DATA ANALYSIS. Everything about R—from its object types to its default printing behaviour—is tailored to the sorts of things data analysts do every day.

By contrast, Python and Julia are general-purpose languages. They are designed to do many things well: build websites, run simulations, automate tasks, and yes, analyse data. But this means that good data analysis is a goal these languages aspire to, not one they were born to achieve.

To illustrate, suppose you wanted to calculate the mean of the numbers 1 to 100. In R, this is as natural as a reflex:

```r
mean(1:100)
```

```
[1] 50.5
```

No need to import packages. No need to loop. No need to define arrays. It just works.

In Python, you will find that a bit more ceremony is required:

```python
x = list(range(1, 101))
print(sum(x) / len(x))
```

```
50.5
```

Additional packages, like Python's excellent *numpy* package can simplify what needs to be written, but it still is not quite so good as what R offers as a baseline user experience.

Julia sits somewhere in the middle. It was built with scientific computing in mind, and its syntax can often be just as clean as R's:

```julia
using Statistics
mean(1:100)
```

However, Julia still expects you to opt in to statistics, with basic functions like `mean()` not being available until you explicitly load them. That is not a flaw BTW—it's a philosophical choice.

R, by contrast, assumes you are doing data analysis. You do not have to ask for permission to compute a mean. Moreover, neither Julia nor Python have the mature ecosystem of statistical tools, diagnostic plots, or nuanced modelling features that R does—at least not yet. R's statistical packages, often written by the very people developing the methods, remain second to none.

That being said Python and Julia are still great choices for data analysis and nobody should be discouraged from using them (to the contrary in fact).

2 How to Use R on a Computer

R will install and run straightforwardly on Windows and Macintosh operating systems as well as Linux;[1] however, prior to attempting any install it is important to make a simple distinction first. R is a programming language, which means it is nothing more than a language you use to communicate instructions to a computer. To communicate those instructions, some type of interface is required. This is a basic reality that applies to any language. It is quite difficult to communicate with someone if they have no mouth, eyes, or ears to send and receive communications with. Computers are no different in this respect. Simply understanding the language is not sufficient. For this reason, most operating systems come equipped with a basic way of interfacing with the user via a terminal emulator which is a windowed application that allows you to type instructions (a.k.a. "commands") to your computer. On computers using the Windows operating system, this is referred to as the *Windows Terminal*, on Macintosh and Linux computers this is the *Terminal* application.[2]

We refer to these as "emulators" because they are emulating the terminals of the past. In a bygone era, "terminals" were peices of computer hardware, consisting mainly of a keyboard paired with a monitor or printer, that people could use to send data to and receive data from a computer. Prior to this, information would have to be entered via punch-card (Norsk Teknisk Museum 2014).

Relying on your operating system's terminal emulator as a primary interface is often a cumbersome and inefficient experience, and definitely not a recommended course of action - though, for what it is worth, Linux users seem to delight in this sort of thing.[3] The preferred means of communicating R to your computer is via the use of what, in programming lingo, is commonly termed an *environment* or, more garrulously, an *integrated development environment* (IDE). This is simply a software application providing the user with a more polished visual workspace and feature set to make programming a smoother experience. IDEs exist for almost every language and for nearly any use case you can imagine. Among R users, RStudio and Positron are two of the most popular options. That said, IDEs vary widely in style and capability, and using R does not require the use of these IDEs specifically, or any IDE for that matter. Still,

[1] Admittedly, the process is probably a little less straightforward on Linux, but still managable.

[2] It's worth noting that most popular mobile and desktop operating systems do their absolute best to keep terminal emulators out of sight. They they offer too much fine grain control of the computer (a.k.a. freedom). That makes it hard to herd users into the unquestioned patterns of behaviour the platform wants. You'll hear this justified as protecting the "user experience," but it rings hollow: 10-year-olds in the early '90s navigated MS-DOS to launch Duke Nukem, and '80s kids taught themselves BASIC to play Ultima. The assumption that users can't handle a command line isn't about safety, it's about control.

[3] I say that as someone who worships at the throne of Tux, the Penguin Emperor.

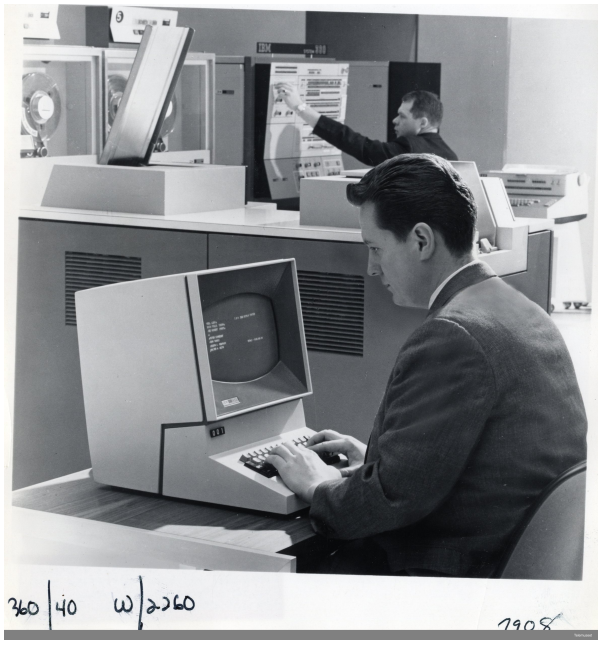


Figure 2.1: IBM 2260 terminal used to communicate with an IBM mainframe computer in the 1960s and 1970s.

installing one alongside the language itself is probably a good idea. However, before doing that, it is important to first install the R language itself.

2.1 Installing the R Programming Language

The standard installation of R will come with an associated environment for the user - provided they are working with either a Windows or Macintosh operating system. An example of the stock environment R ships with on Windows can be seen in Figure 2.2. Admittedly, as IDEs go, this will get the job done, but offers a pretty lackluster experiance. But don't worry, that's what RStudio and Positron are for.

To install R - both the language and environment simultaneously - simply go to the *R Project for Statistical Computing* website: https://www.r-project.org/ The site makes no apologies for its spartan '90s aesthetic, and we should all be so principled. Zero flash, maximum substance.

Located on the front-page of this website should be a link labelled CRAN. This stands for "Comprehensive R Archive Network" and is a set of servers around the world that distribute R alongside packages associated with R. The servers are "mirrored," " meaning they all provide the same content. So there is no need to worry about one server providing incomplete, out-of-date, or unofficial versions of R. Technically speaking, the server closest to your home location is the

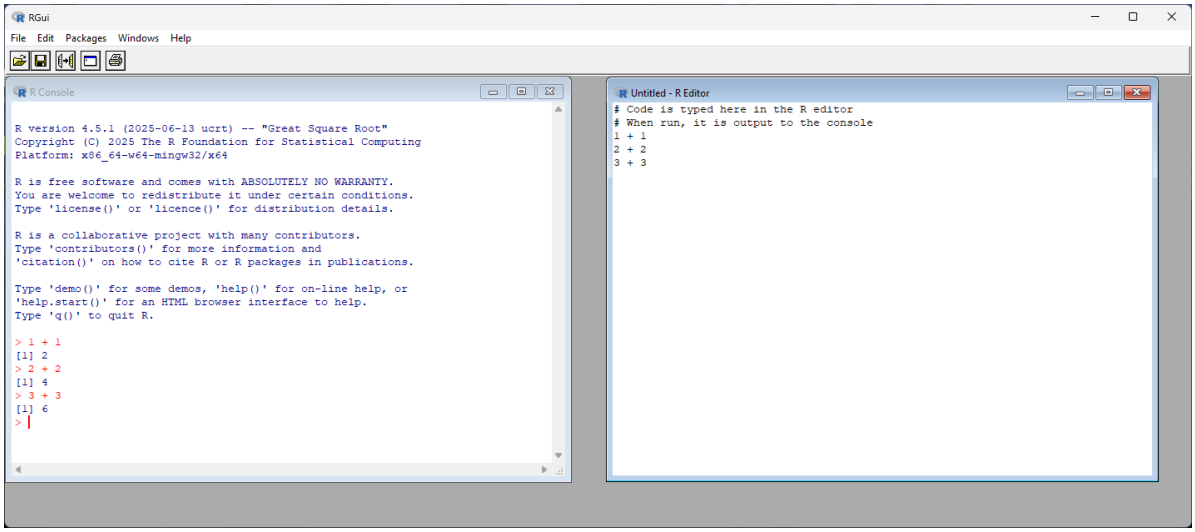


Figure 2.2: Base R environment for Windows, featuring the default console interface for running R code and viewing outputs (left) and the default scripting window (right) for writing, running, and saving R code.

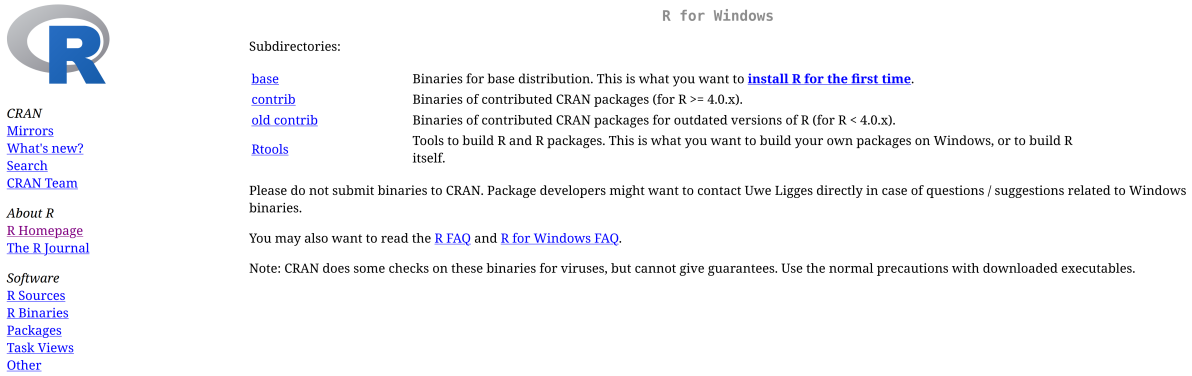
one you should opt to download from; however, the topmost link labeled "0-Cloud" will be sufficient for most users. The install file is only about 80 megabytes large, so unless you live in the remotest areas of Earth, download speed, and thus choice of server, is probably not a concern.

Once you have chosen a suitable server, you will need select your operating system from the list the site provides (Chromebooks and Mobile devices are out of luck - though cloud-based alternatives do exist that can run R through a web-browser).

2.1.1 Windows Installation

Click the "Download for Windows" link to access the R download page(see Figure 2.3). Select the **"base"** option (labeled "install R for the first time"), which will direct you to the installer for the latest version of R. You'll see a download link for the current release—for example, **Download R-4.5.3 for Windows** - which is the newest version available at the time of writing.

Once you have downloaded the installation file, open it and follow the setup wizard. Keep the default settings unless you have a specific reason to change them. Once installation is complete, you can launch R from the Start menu or a desktop shortcut (if you chose to create one during setup).



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Subdirectories:

base	Binaries for base distribution. This is what you want to **install R for the first time**.
contrib	Binaries of contributed CRAN packages (for R >= 4.0.x).
old contrib	Binaries of contributed CRAN packages for outdated versions of R (for R < 4.0.x).
Rtools	Tools to build R and R packages. This is what you want to build your own packages on Windows, or to build R itself.

Please do not submit binaries to CRAN. Package developers might want to contact Uwe Ligges directly in case of questions / suggestions related to Windows binaries.

You may also want to read the R FAQ and R for Windows FAQ.

Note: CRAN does some checks on these binaries for viruses, but cannot give guarantees. Use the normal precautions with downloaded executables.

Figure 2.3: Windows download options

Coming Soon:

- Installation Instructions for R, RStudio, Positron on Windows, Mac, and (maybe) beginner friendly Linux Distros.
- Cloud-based IDEs (no installation required)

References

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